Guidelines for collecting and processing gene expression data with XpBrew

- Strictly use the following filters for selecting the SRA files from NCBI:
    - Click on the species name in the taxonomy box on the right of the page to restrict only the specific species samples
    - Source: RNA
    - Library: Paired
    - Platform: Illumina, BGISeq
    - File type: fastq

- Please make sure to choose the best available annotation for the species when you collect its SRA files for making the expression file; Please also make sure that they are not restricted in any way. RESTRICTED DATASETS from any source should be AVOIDED
- Jakob has an annotation collection of ~2000 angiosperms from NCBI of mostly very good quality - so make use of that and go for other sources only if you are not able to find your plant species in our in-house collections
- Ensure that the CDS FASTA headers correspond to the transcript attributes in the GFF file to facilitate smooth downstream steps that need GFF files and the CDS file headers to sync, like synteny analyses; I recommend using the get_cds_from_assembly.py helper script found in the XpBrew repo for this. In case you want to compare the generated CDS sequences with CDS sequences from any other source, please use the compare_cds_seqs.py for the same (it needs seqkit2 to be installed). Check the headers of the CDS file after they are generated to ensure that they have the required transcript feature attribute in place.
- From XpBrew_v0.5.1 onwards metadata collection is being enforced as a part of the pipeline i.e. you will be able to use the pipeline only if you provide the required metadata config file in the proper format. Refer to the pipeline usage section for more details.
- An additional pointer on taking references - if you are not able to find supporting publication DOIs for a dataset from NCBI/ GWH use the bioproject link instead. In case of phytozome you will be able to find the citation link in the info page of that species that can be accessed with the tiny i symbol near the species name on the Phytozome home page

Guidelines to submit data for PanXpresso collection

- There is a storage VM being maintained as a central node for all expression collection datasets.
- Before you transfer the folder with the required expression collection data of the species, ensure that the name of the folder is the same as the species name that you have provided in the mandatory --sample_name flag when using the pipeline
- Every folder must have 4 files mandatorily - genus_species.cds.fasta, genus_species.pep.fasta, genus_species.tpms.tsv, genus_species.json, and if isoform removal succeeded then 7 files in total including genus_species.repr.cds.fasta,

genus_species.repr.pep.fasta, and genus_species.repr.tpms.tsv; The json file is important as it will be used by an automated script that scouts the storage folders' for json files to update the metadata in a tsv file that gets regularly synced with the metadata tracker sheet in Sciebo in the path /PBB/Projects/PanXpresso_tracker

- If you are submitting datasets for PanXpresso - there are 3 kinds of folders in which you can add your data - Re_annotation_plants, New_addition_plants and Extension_plants
  -> Re_annotation concerns the use of an improved or better annotation for a plant species that already exists in PanXpresso
  ->Extension involves extending the number of SRA samples with an existing annotation for a plant species already available in PanXpresso
  -> New_addition_plants concerns addition of datasets for plant species that are not yet there in the PanXpresso collection
  Deposit your folders into one of these three folders according to the use-case, and an automated script will scout these folders at regular intervals to add them to the main PanXpresso plants collection